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(54) **AUGMENTED-REALITY SURGICAL SYSTEM USING DEPTH SENSING**

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See application file for complete search history.

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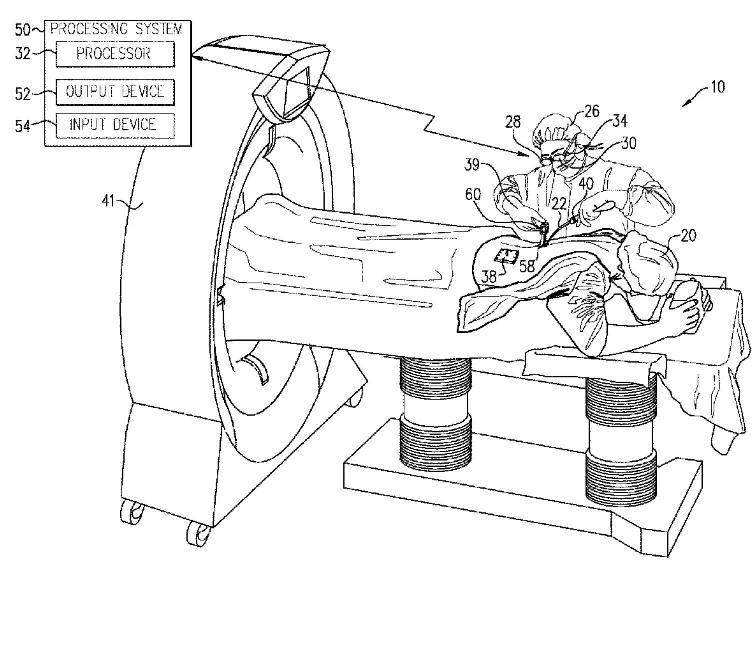
(57) **ABSTRACT**

Disclosed herein are systems, devices, and methods for image-guided surgery. Some systems include a head-mounted unit, having a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) of a body of a patient that is viewed through the display by a user wearing the head-mounted unit. A processor, which is configured to receive a three-dimensional (3D) tomographic image of the body of the patient, computes a depth map of the ROI based on the depth data generated by the depth sensor, to compute a transformation over the ROI so as to register the tomographic image with the depth map, and to apply the transformation in presenting a part of the tomographic image on the display in registration with the ROI viewed through the display.

20 Claims, 9 Drawing Sheets



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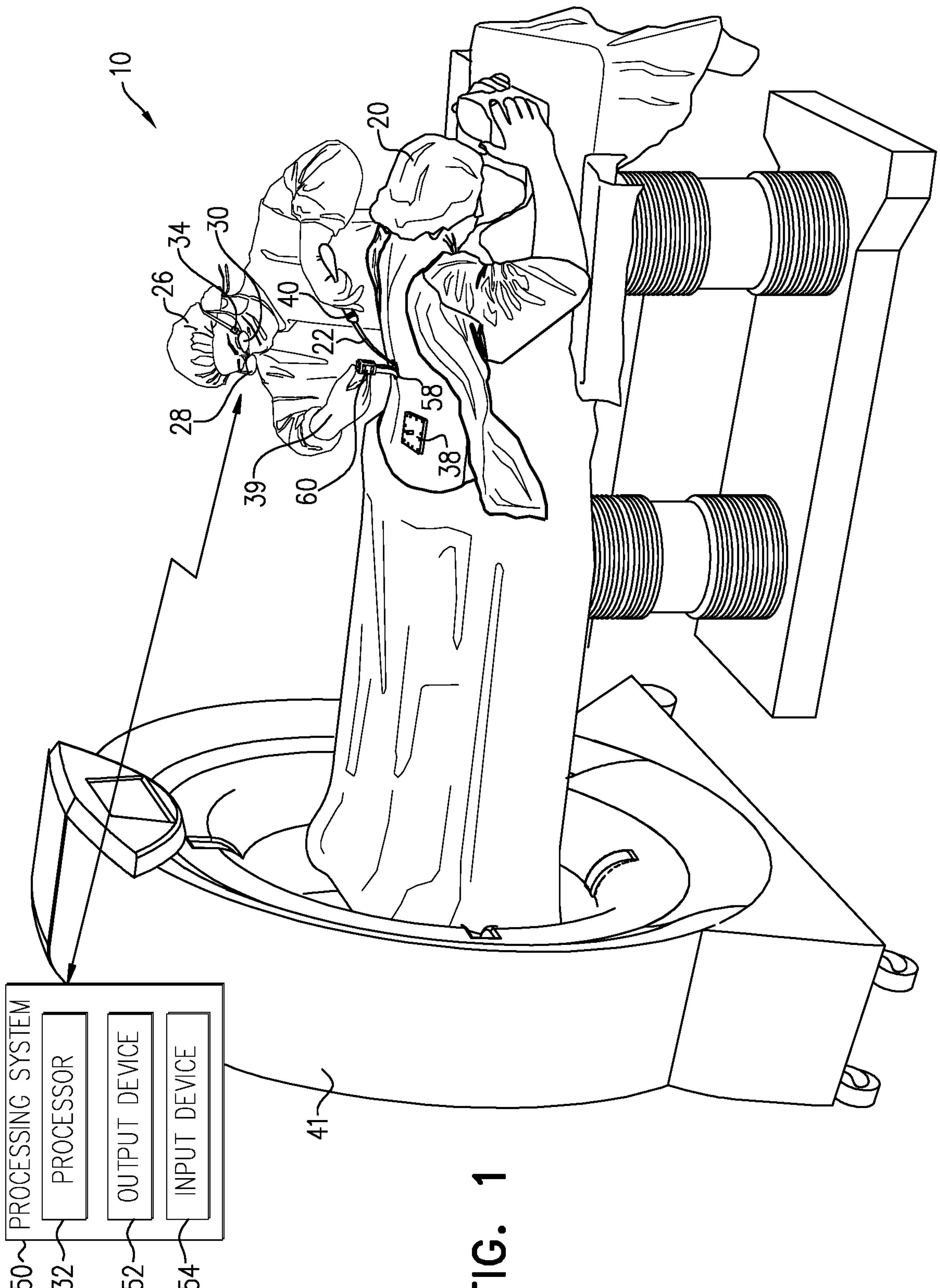
10
20
34
30
26
40
22
28
58
38
39
60
41
PROCESSING SYSTEM
PROCESSOR
OUTPUT DEVICE
INPUT DEVICE
50
32
52
54

FIG. 1

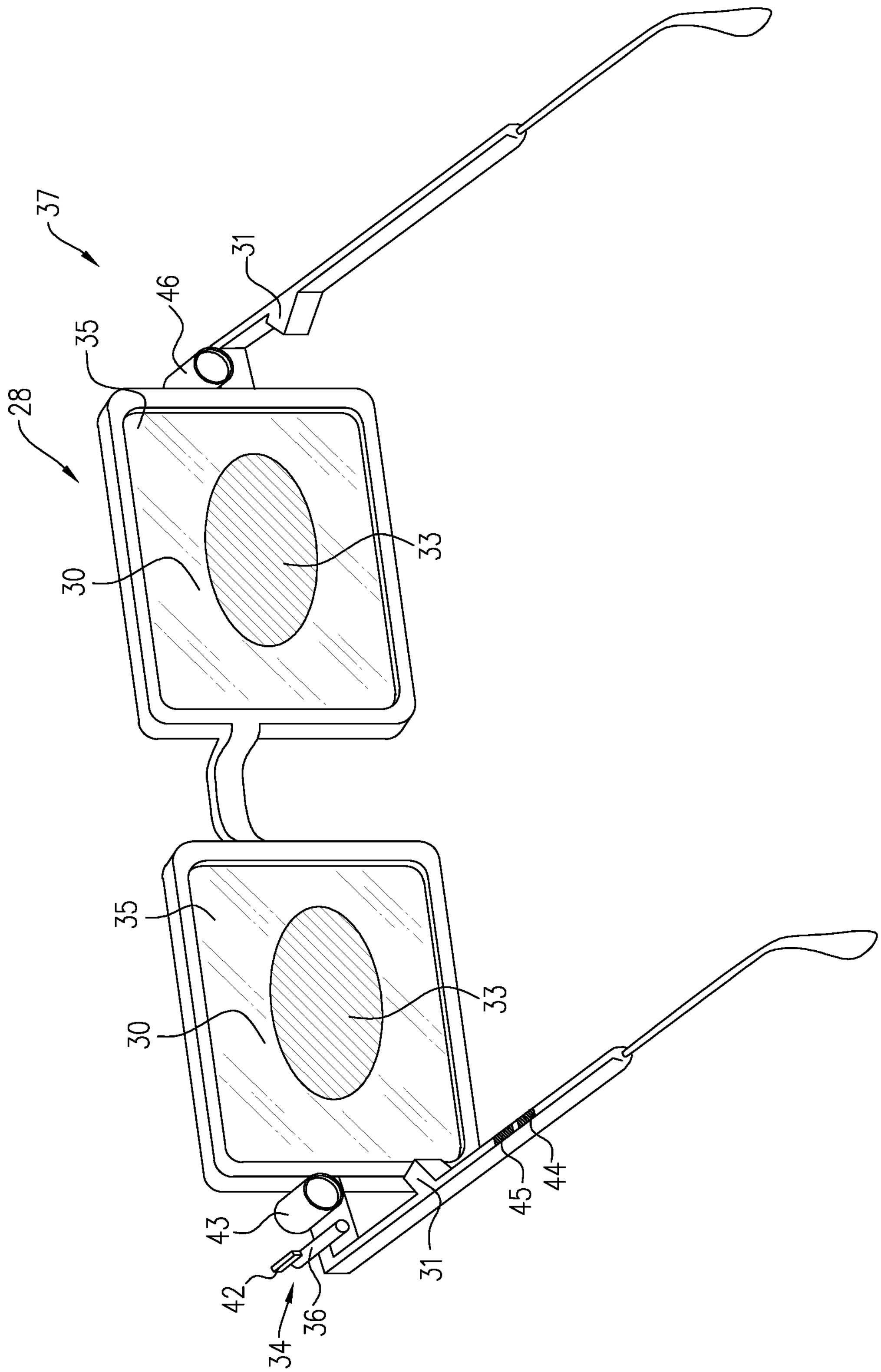

FIG. 2A

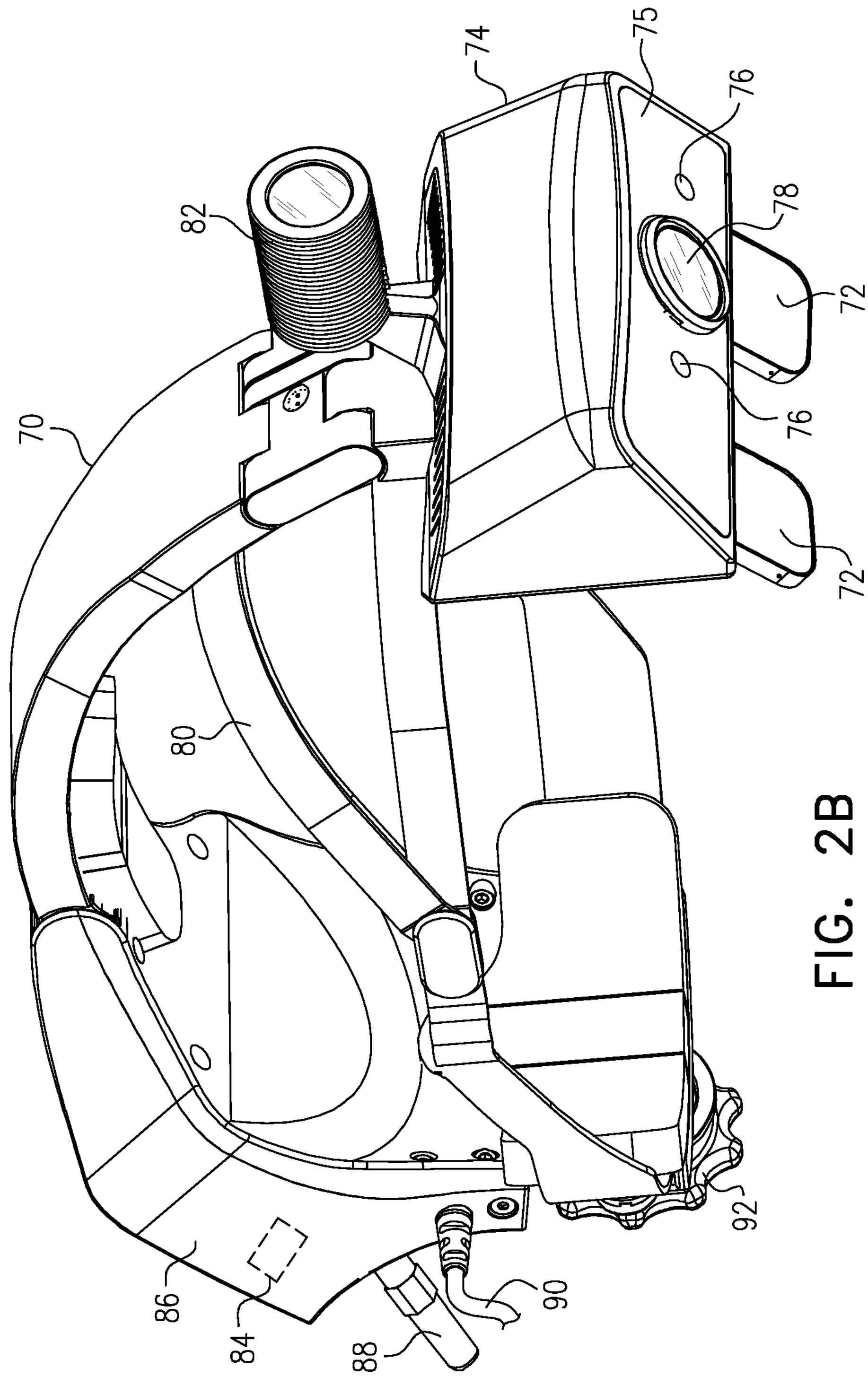

FIG. 2B

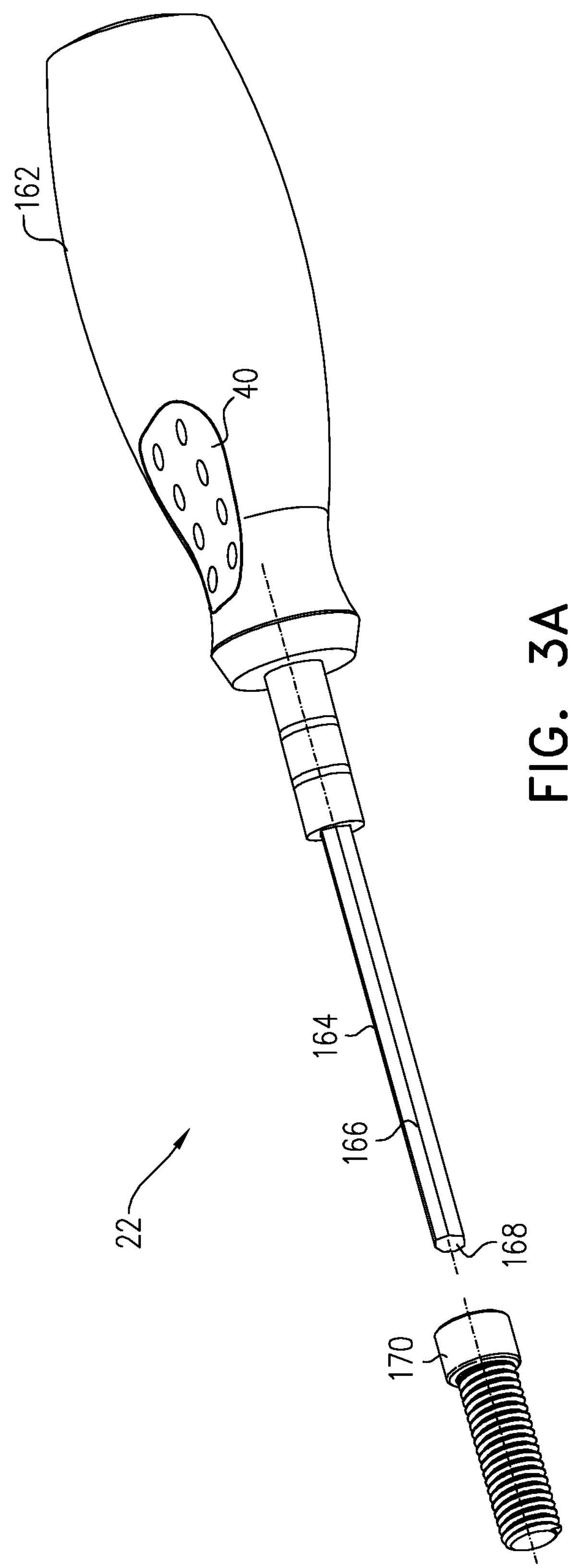

FIG. 3A

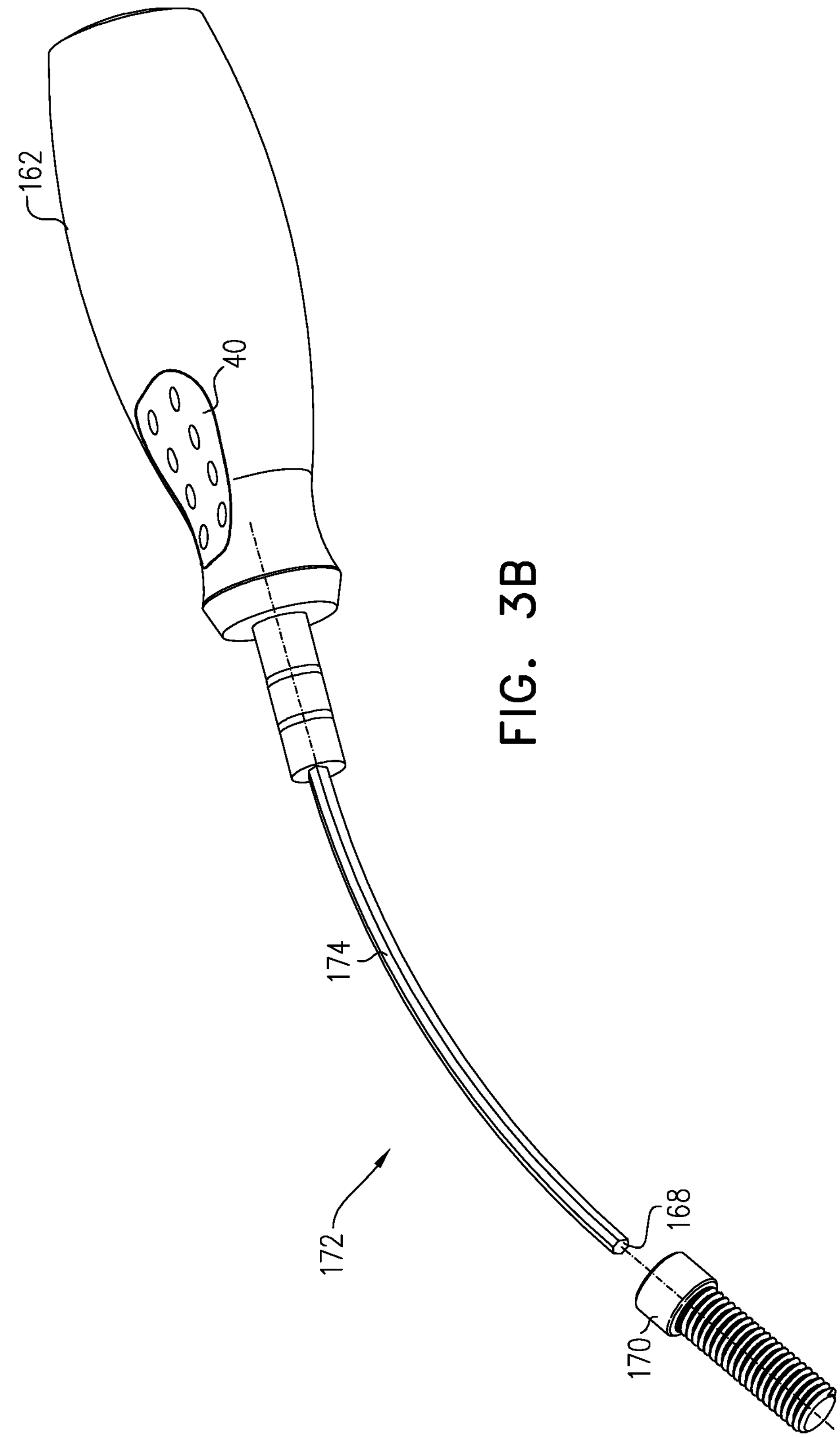

FIG. 3B

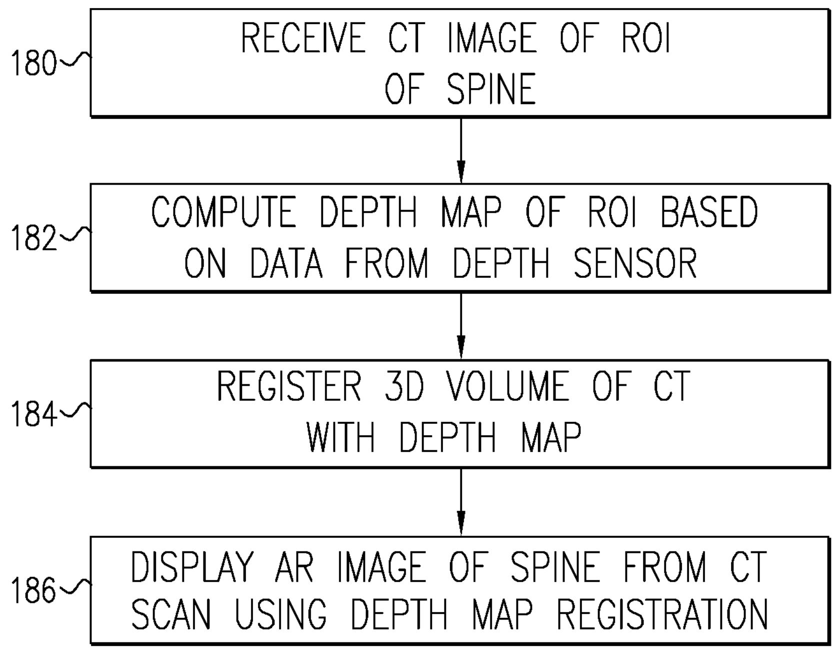
180
RECEIVE CT IMAGE OF ROI OF SPINE
182
COMPUTE DEPTH MAP OF ROI BASED ON DATA FROM DEPTH SENSOR
184
REGISTER 3D VOLUME OF CT WITH DEPTH MAP
186
DISPLAY AR IMAGE OF SPINE FROM CT SCAN USING DEPTH MAP REGISTRATION

FIG. 4A

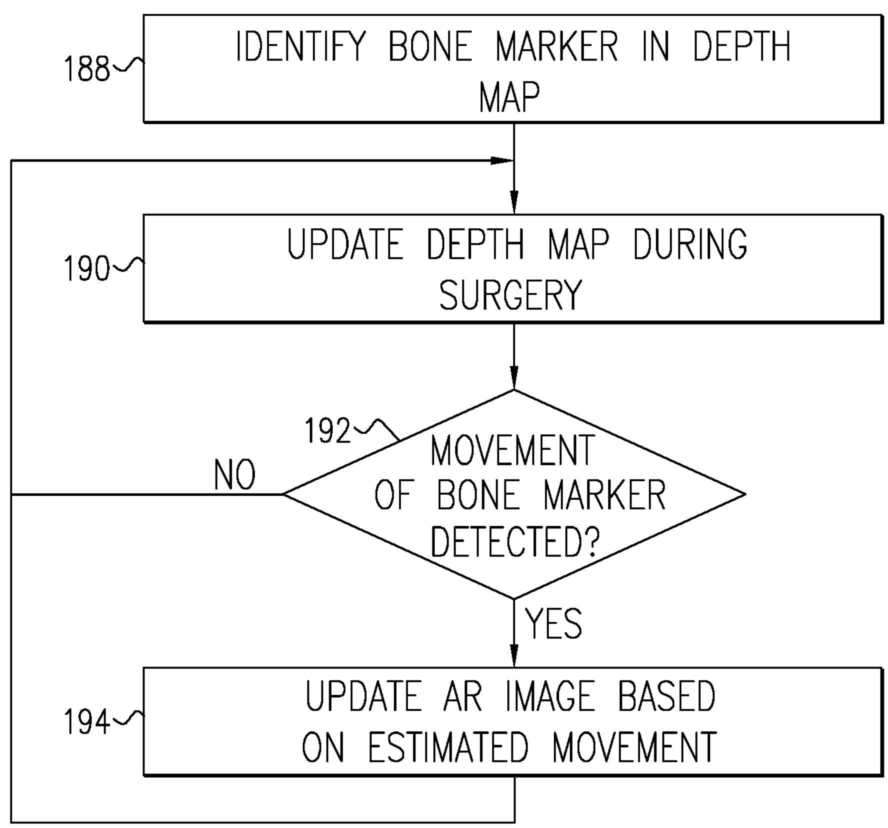
188
IDENTIFY BONE MARKER IN DEPTH MAP
190
UPDATE DEPTH MAP DURING SURGERY
192
MOVEMENT OF BONE MARKER DETECTED?
NO
YES
194
UPDATE AR IMAGE BASED ON ESTIMATED MOVEMENT

FIG. 4B

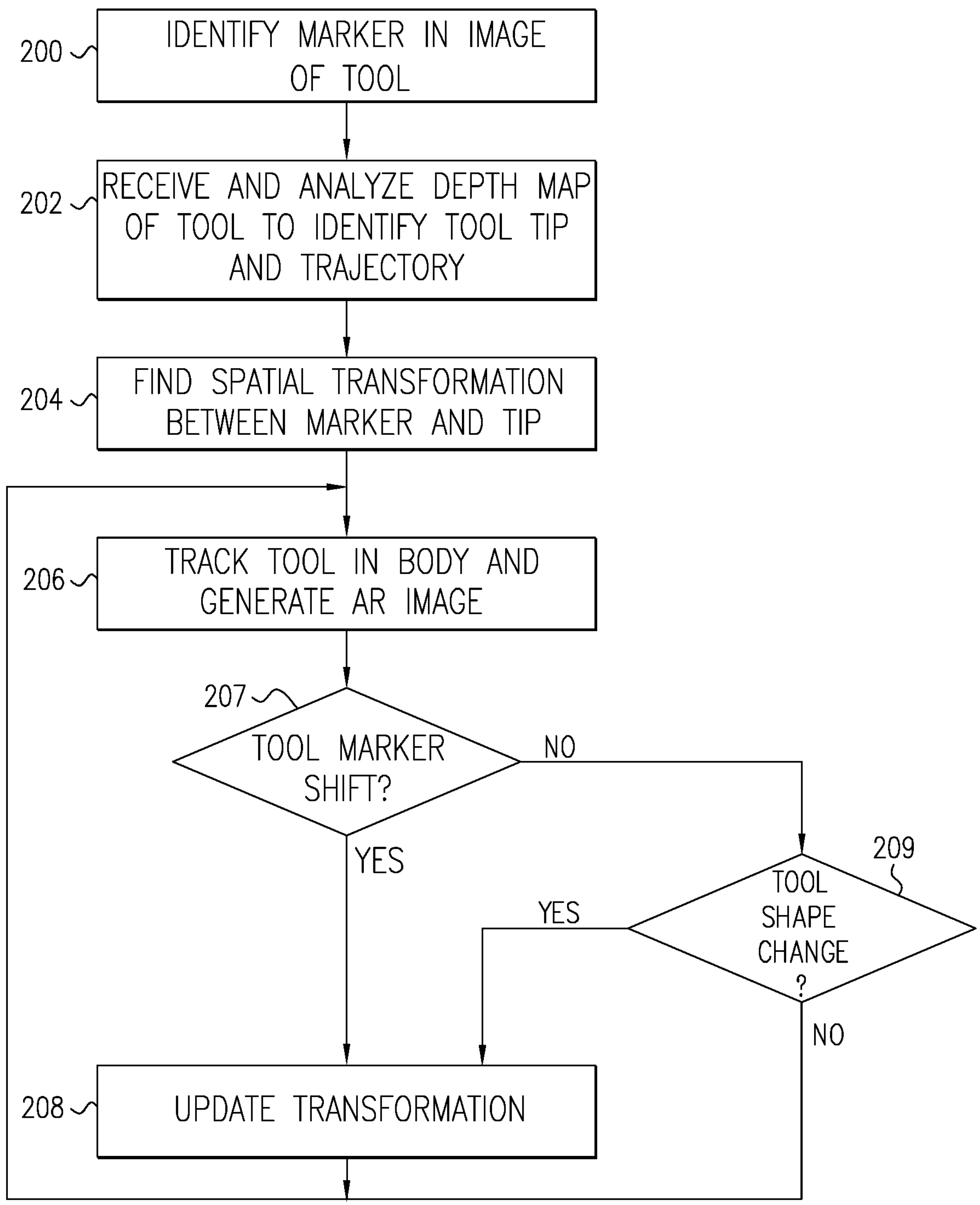
200
IDENTIFY MARKER IN IMAGE OF TOOL
202
RECEIVE AND ANALYZE DEPTH MAP OF TOOL TO IDENTIFY TOOL TIP AND TRAJECTORY
204
FIND SPATIAL TRANSFORMATION BETWEEN MARKER AND TIP
206
TRACK TOOL IN BODY AND GENERATE AR IMAGE
207
TOOL MARKER SHIFT?
NO
YES
209
TOOL SHAPE CHANGE ?
YES
NO
208
UPDATE TRANSFORMATION

FIG. 5

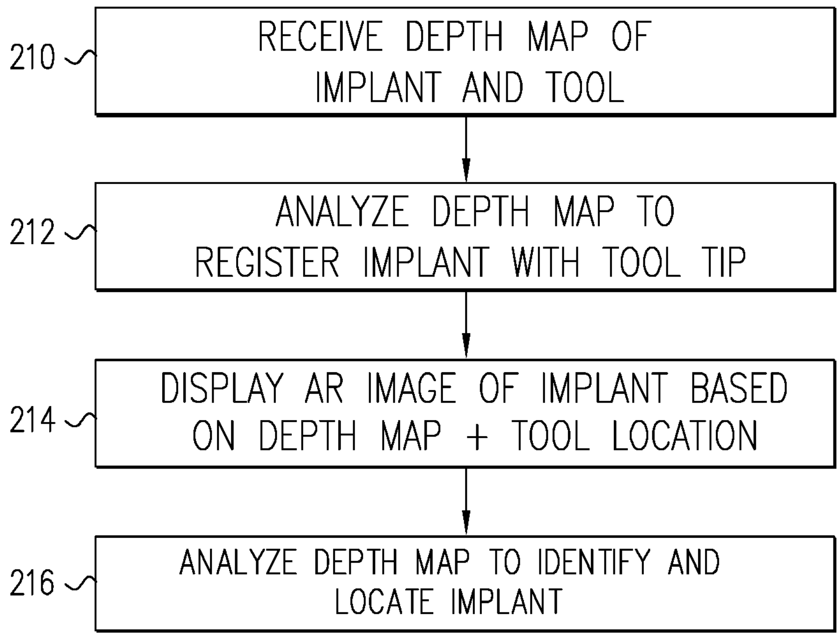
210
RECEIVE DEPTH MAP OF IMPLANT AND TOOL
212
ANALYZE DEPTH MAP TO REGISTER IMPLANT WITH TOOL TIP
214
DISPLAY AR IMAGE OF IMPLANT BASED ON DEPTH MAP + TOOL LOCATION
216
ANALYZE DEPTH MAP TO IDENTIFY AND LOCATE IMPLANT

FIG. 6

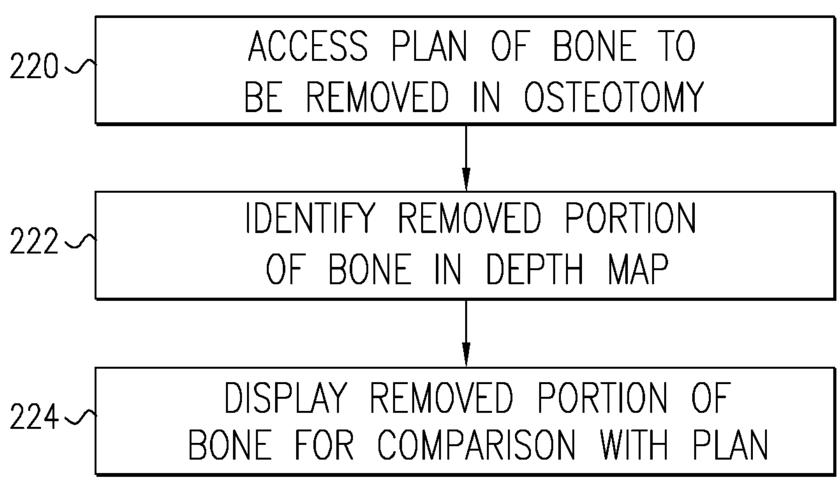
220
ACCESS PLAN OF BONE TO BE REMOVED IN OSTEOTOMY
222
IDENTIFY REMOVED PORTION OF BONE IN DEPTH MAP
224
DISPLAY REMOVED PORTION OF BONE FOR COMPARISON WITH PLAN

FIG. 7

FIG. 8

AUGMENTED-REALITY SURGICAL SYSTEM USING DEPTH SENSING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International PCT Application PCT/IB2022/057733, filed Aug. 18, 2022, which claims the benefit of: U.S. Provisional Patent Application 63/236,241, filed Aug. 24, 2021; U.S. Provisional Patent Application 63/281,677, filed Nov. 21, 2021; U.S. Provisional Patent Application No. 63/234,272, filed Aug. 18, 2021; and U.S. Provisional Patent Application No. 63/236,244, filed Aug. 24, 2021. The entire content of each of these related applications is incorporated herein by reference.

FIELD

The present disclosure relates generally to image-guided surgery or intervention, and specifically to systems and methods for use of augmented reality in image-guided surgery or intervention and/or to systems and methods for use in surgical computer-assisted navigation.

BACKGROUND

Near-eye display devices and systems can be used in augmented reality systems, for example, for performing image-guided surgery. In this way, a computer-generated image may be presented to a healthcare professional who is performing the procedure such that the image is aligned with an anatomical portion of a patient who is undergoing the procedure. Applicant's own work has demonstrated an image of a tool that is used to perform the procedure can also be incorporated into the image that is presented on the head-mounted display. For example, Applicant's prior systems for image-guided surgery have been effective in tracking the positions of the patient's body and the tool (see, for example, U.S. Pat. Nos. 9,928,629, 10,835,296, 10,939,977, PCT International Publication WO 2019/211741, and U.S. Patent Application publication 2020/0163723.) The disclosures of all these patents and publications are incorporated herein by reference.

SUMMARY

Embodiments of the present disclosure provide improved systems, methods, and software for image-guided surgery. Some embodiments of the systems improve the accuracy of the augmented-reality images that are presented on the display and broaden the capabilities of the augmented reality system by employing depth sensing.

In some embodiments, a system for image-guided surgery comprises a head-mounted unit, comprising a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) of a body of a patient that is viewed through the display by a user wearing the head-mounted unit; and a processor, which is configured to receive a three-dimensional (3D) tomographic image of the body of the patient, to compute a depth map of the ROI based on the depth data generated by the depth sensor, to compute a transformation over the ROI so as to register the tomographic image with the depth map, and to apply the transformation in presenting a part of the tomographic image on the display in registration with the ROI viewed through the display.

In some embodiments, the ROI comprises a bone of the body to which an anchoring device is fastened, and wherein the processor is further configured to identify a location of the anchoring device in the depth map, to update the depth map in the course of a surgery, to detect a change in the location of the anchoring device in the updated depth map, and to take a corrective action responsively to the change.

In some embodiments, the corrective action comprises modifying a presentation on the display responsively to the change in the location of the anchoring device.

In some embodiments, the depth map includes a spine of the patient, which is exposed in a surgical procedure, and wherein the processor is configured to compute the transformation by registering the spine in the depth map with the spine appearing in the tomographic image.

In some embodiments, the processor is configured to process the depth data so as to detect a position of a marker that is fixed to the body of the patient, to recognize a location of the head-mounted unit relative to the body based on the detected position, and to position the image presented on the display responsively to the recognized location.

In some embodiments, the processor is configured to process the depth data so as to identify a change in an anatomical structure in the body of the patient during a surgical procedure, and to modify the image presented on the display responsively to the identified change.

In some embodiments, the processor is configured to process the depth data so as to identify an implant inserted into the body of the patient during a surgical procedure, and to modify the image presented on the display responsively to the identified implant.

In some embodiments, the tomographic image comprises a CT scan of the patient, which was performed with an array of radiopaque fiducial markers fixed to the body of the patient, and wherein the processor is configured to identify respective 3D coordinates of the fiducial markers in the depth map and to register the CT scan with the ROI viewed through the display by matching the fiducial markers in the CT to the respective 3D coordinates.

In some embodiments, a system for image-guided surgery comprises a head-mounted unit, comprising: a see-through augmented-reality display; and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit and with respect to a surgical tool when the tool is placed within a field of view of the depth sensor, wherein the tool comprises a shaft and a marker containing a predefined pattern disposed on the tool in a fixed spatial relation to the shaft; and a processor, which is configured to: process the depth data so as to identify a shape of the tool and to compute, responsively to the shape, a spatial transformation between a position of the marker and a location and orientation of the shaft; track the position of the marker as the user manipulates the shaft of the tool within the body, and using the tracked position and the spatial transformation, generate an image of the tool, including the shaft, on the display in registration with the ROI viewed through the display.

In some embodiments, further comprising a tracking sensor, which is disposed on the head-mounted unit in a known spatial relation to the depth sensor and is configured to detect the position of the marker.

In some embodiments, the shaft has a curved shape, and wherein the processor is configured to process the depth data so as to reconstruct a three-dimensional (3D) model of the curved shape, and to generate the image of the tool based on the 3D model.

In some embodiments, the processor is configured to process the depth data so as detect a change in a shape of the tool and to update the image of the tool on the display responsively to the change in the shape.

In some embodiments, the depth sensor is further configured to generate the depth data with respect to a further marker that is attached to the body of the patient, and wherein the processor is configured to apply the depth data in calculating a position of the tool relative to the body.

In some embodiments, the depth sensor is configured to generate further depth data with respect to a hand of a user of the head-mounted unit, and wherein the processor is configured to process the further depth data so as to detect a gesture made by the hand, and to control a function of the system responsively to the detected gesture.

In some embodiments, a system for image-guided surgery, comprises a head-mounted unit, comprising a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit and with respect to a surgical implant when the implant is placed within a field of view of the depth sensor, wherein the implant is configured to be mounted on a shaft of a surgical tool and inserted, using the tool, into the body; and a processor, which is configured to process the depth data so as to identify a shape of the implant and to compute, responsively to the shape, a spatial transformation between a position of a marker disposed on the tool and a location and orientation of the implant, to track the position of the marker as the user manipulates the shaft of the tool within the body, and using the tracked position, the spatial transformation, and the identified shape, to generate on the display an image of the implant within the body in registration with the ROI viewed through the display, wherein the marker contains a predefined pattern and is disposed in a fixed spatial relation to the shaft.

In some embodiments, the system further comprising a tracking sensor, which is disposed on the head-mounted unit in a known spatial relation to the depth sensor and is configured to detect the position of the marker.

In some embodiments, the shaft has a curved shape, and wherein the processor is configured to process the depth data so as to reconstruct a three-dimensional (3D) model of the curved shape, and to generate the image of the implant based on the 3D model.

In some embodiments, the processor is configured to process the depth data so as detect a change in a shape of the tool and to update the spatial transformation responsively to the change in the shape.

In some embodiments, a system for image-guided surgery, comprises a head-mounted unit, comprising a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) on a body of a patient, including a bone inside the body, that is viewed through the display by a user wearing the head-mounted unit; and a processor, which is configured to process the depth data generated by the depth sensor so as to identify a first three-dimensional (3D) shape of the bone prior to a surgical procedure on the bone and a second 3D shape of the bone following the surgical procedure, and to generate, based on the first and second 3D shapes, an image showing a part of the bone that was removed in the surgical procedure.

In some embodiments, a method for image-guided surgery comprises using a head-mounted unit that includes a see-through augmented-reality display and a depth sensor, generating depth data with respect to a region of interest (ROI) of a body of a patient that is viewed through the display by a user wearing the head-mounted unit; receiving a three-dimensional (3D) tomographic image of the body of the patient; computing a depth map of the ROI based on the depth data generated by the depth sensor; computing a transformation over the ROI so as to register the tomographic image with the depth map; and applying the transformation in presenting a part of the tomographic image on the display in registration with the ROI viewed through the display.

In some embodiments, the ROI comprises a bone of the body to which an anchoring device is fastened, and wherein the method comprises: identifying an initial location of the anchoring device in the depth map; updating the depth map in the course of a surgery; detecting a change in the location of the anchoring device in the updated depth map; and taking a corrective action responsively to the change.

In some embodiments, taking the corrective action comprises modifying a presentation on the display responsively to the change in the location of the anchoring device.

In some embodiments, the depth map includes a spine of the patient, which is exposed in a surgical procedure, and wherein computing the transformation comprises registering the spine in the depth map with the spine appearing in the tomographic image.

In some embodiments, the method further comprises processing the depth data so as to detect a position of a marker that is fixed to the body of the patient; recognizing a location of the head-mounted unit relative to the body based on the detected position; and positioning the image presented on the display responsively to the recognized location.

In some embodiments, the method further comprises processing the depth data so as to identify a change in an anatomical structure in the body of the patient during a surgical procedure; and modifying the image presented on the display responsively to the identified change.

In some embodiments, the method further comprises processing the depth data so as to identify an implant inserted into the body of the patient during a surgical procedure; and modifying the image presented on the display responsively to the identified implant.

In some embodiments, the tomographic image comprises a CT scan of the patient, which was performed with an array of radiopaque fiducial markers fixed to the body of the patient, and wherein computing the transformation comprises identifying respective 3D coordinates of the fiducial markers in the depth map, and registering the CT scan with the ROI viewed through the display by matching the fiducial markers in the CT to the respective 3D coordinates.

In some embodiments, a method for image-guided surgery comprises using a head-mounted unit that includes a see-through augmented-reality display and a depth sensor, generating depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit and with respect to a surgical tool when the tool is placed within a field of view of the depth sensor, wherein the tool comprises a shaft and a marker containing a predefined pattern disposed on the tool in a fixed spatial relation to the shaft; processing the depth data so as to identify a shape of the tool and to compute, responsively to the shape, a spatial transformation between a position of the marker and a location and orientation of the shaft; tracking the position of the marker as the user manipulates the shaft of the tool within the body; and using the tracked position and the spatial transformation, generating an image of the tool, including the shaft, on the display in registration with the ROI viewed through the display.

In some embodiments, the tracking the position comprises detecting the position of the marker using a tracking sensor disposed on the head-mounted unit in a known spatial relation to the depth sensor.

In some embodiments, the shaft has a curved shape, and wherein processing the depth data comprises reconstructing a three-dimensional (3D) model of the curved shape, wherein the image of the tool is generated based on the 3D model.

In some embodiments, processing the depth data comprises detecting a change in a shape of the tool, and wherein generating the image comprises updating the image of the tool on the display responsively to the change in the shape.

In some embodiments, generating the depth data comprises capturing further depth data with respect to a further marker that is attached to the body of the patient, and wherein processing the depth data comprises applying the further depth data in calculating a position of the tool relative to the body.

In some embodiments, generating the depth data comprises capturing further depth data with respect to a hand of a user of the head-mounted unit, and wherein the method comprises processing the further depth data so as to detect a gesture made by the hand, and controlling a function of the head-mounted unit responsively to the detected gesture.

In some embodiments, a method for image-guided surgery comprises using a head-mounted unit that includes a see-through augmented-reality display and a depth sensor, generating depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit and with respect to a surgical implant when the implant is placed within a field of view of the depth sensor, wherein the implant is mounted on a shaft of a surgical tool and inserted, using the tool, into the body; processing the depth data so as to identify a shape of the implant; computing, responsively to the shape, a spatial transformation between a position of a marker disposed on the tool and a location and orientation of the implant, wherein the marker contains a predefined pattern and is disposed in a fixed spatial relation to the shaft; tracking the position of the marker as the user manipulates the shaft of the tool within the body; and using the tracked position, the spatial transformation, and the identified shape, generating on the display an image of the implant within the body in registration with the ROI viewed through the display.

In some embodiments, the method further comprises detecting the position of the marker using a tracking sensor, which is disposed on the head-mounted unit in a known spatial relation to the depth sensor.

In some embodiments, the shaft has a curved shape, and wherein processing the depth data comprises reconstructing a three-dimensional (3D) model of the curved shape, wherein the image of the implant is generated based on the 3D model.

In some embodiments, processing the depth data comprises detecting a change in a shape of the tool, and updating the spatial transformation responsively to the change in the shape.

In some embodiments, a method for image-guided surgery, comprises using a head-mounted unit that includes a see-through augmented-reality display and a depth sensor, generating depth data with respect to a region of interest (ROI) on a body of a patient, including a bone inside the body, that is viewed through the display by a user wearing the head-mounted unit; processing the depth data generated by the depth sensor so as to identify a first three-dimensional (3D) shape of the bone prior to a surgical procedure on the bone and a second 3D shape of the bone following the surgical procedure; and generating, based on the first and second 3D shapes, an image showing a part of the bone that was removed in the surgical procedure.

In some embodiments, the surgical procedure involves a bone cut.

In some embodiments, a head-mounted system for image-guided surgery comprises a see-through augmented-reality display disposed so as to be viewable by a user over a region of interest (ROI) of a body of a patient; a depth sensor configured to generate depth data with respect to the ROI; and a processor and a memory for storing instructions that, when executed by the processor cause the system to: receive a three-dimensional (3D) tomographic image of the body of the patient; determine a depth map of the ROI based at least in part on the depth data; determine a transformation over the ROI so as to register the 3D tomographic image with the depth map; and display at least a part of the 3D tomographic image on the see-through augmented-reality display in registration with the ROI based at least in part on the transformation.

In some embodiments, a method for image-guided surgery comprises a see-through augmented-reality display disposed so as to be viewable by a user over a region of interest (ROI) of a body of a patient; a depth sensor configured to generate depth data with respect to the ROI; and a processor and a memory for storing instructions that, when executed by the processor cause the system to: receive a three-dimensional (3D) tomographic image of the body of the patient; determine a depth map of the ROI based at least in part on the depth data; determine a transformation over the ROI so as to register the 3D tomographic image with the depth map; and display at least a part of the 3D tomographic image on the see-through augmented-reality display in registration with the ROI based at least in part on the transformation.

For purposes of summarizing the disclosure, certain aspects, advantages, and novel features are discussed herein. It is to be understood that not necessarily all such aspects, advantages, or features will be embodied in any particular embodiment of the disclosure, and an artisan would recognize from the disclosure herein a myriad of combinations of such aspects, advantages, or features.

The present invention will be more fully understood from the following detailed description of the embodiments thereof, taken together with the drawings in which:

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting features of some embodiments of the invention are set forth with particularity in the claims that follow. The following drawings are for illustrative purposes only and show non-limiting embodiments. Features from different figures may be combined in several embodiments.

FIG. **1** is a schematic pictorial illustration showing a system for image-guided surgery, in accordance with an embodiment of the disclosure;

FIG. **2A** is a schematic pictorial illustration showing details of a near-eye unit that is used for image-guided surgery, in accordance with an embodiment of the disclosure;

FIG. 2B is a schematic pictorial illustration showing details of a head-mounted unit that is used for image-guided surgery, in accordance with another embodiment of the disclosure;

FIG. 3A is a schematic pictorial illustration showing details of a surgical tool, in accordance with an embodiment of the disclosure;

FIG. 3B is a schematic pictorial illustration showing details of a surgical tool, in accordance with another embodiment of the disclosure;

FIGS. 4A and 4B are flow charts that schematically illustrate methods for image-guided surgery, in accordance with embodiments of the disclosure;

FIGS. 5, 6 and 7 are flow charts that schematically illustrate methods for image-guided surgery, in accordance with further embodiments of the disclosure; and

FIG. 8 is a schematic pictorial illustration showing a system for image-guided surgery, in accordance with an alternative embodiment of the disclosure.

DETAILED DESCRIPTION

Overview

Embodiments of the present disclosure that are described herein provide systems, methods and software for image-guided surgery or other intervention, computer assisted navigation and/or stereotactic surgery or other intervention that, inter alia, use depth sensing to enhance the capabilities of an augmented-reality display and system. In some embodiments, a head-mounted unit comprises both a see-through augmented-reality display and a depth sensor. In some embodiments, the depth sensor generates depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit. In some embodiments, the system applies the depth data in generating one or more depth maps of the body. Additionally or alternatively, the depth sensor may be applied in generating depth data with respect to implements such as clamps, tools and implants that can be inserted into the body. In some embodiments, using the depth data, the system is able to improve the accuracy of the augmented-reality images that are presented on the display and broaden the capabilities of the augmented-reality system.

In some embodiments, the term "depth sensor" refers to one or more optical components that are configured to capture a depth map of a scene. For example, in some embodiments, the depth sensor can be a pattern projector and a camera for purposes of structured-light depth mapping. For example, in some embodiments, the depth sensor can be a pair of cameras configured for stereoscopic depth mapping. For example, in some embodiments, the depth sensor can be a beam projector and a detector (or an array of detectors) configured for time-of-flight measurement. Of course the term "depth sensor" as used herein is not limited to the listed examples and can other structure.

System Description

Reference is now made to FIGS. 1 and 2A, which schematically illustrate an exemplary system 10 for image-guided surgery, in accordance with some embodiments of the disclosure. For example, FIG. 1 is a pictorial illustration of the system 10 as a whole, while FIG. 2A is a pictorial illustration of a near-eye unit that is used in the system 10. The near eye unit illustrated in FIGS. 1 and 2A is configured as a head-mounted unit 28. In some embodiments, the near-eye unit can be configured as the head-mounted unit 28 shown in FIG. 8 and as a head-mounted AR display (HMD) unit 70, 100 and described hereinbelow. In FIG. 1, the system 10 is applied in a medical procedure on a patient 20 using image-guided surgery. In this procedure, a tool 22 is inserted via an incision in the patient's back in order to perform a surgical intervention. Alternatively, the system 10 and the techniques described herein may be used, mutatis mutandis, in other surgical procedures.

Methods for optical depth mapping can generate a three-dimensional (3D) profile of the surface of a scene by processing optical radiation reflected from the scene. In the context of the present description and in the claims, the terms depth map, 3D profile, and 3D image are used interchangeably to refer to an electronic image in which the pixels contain values of depth or distance from a reference point, instead of or in addition to values of optical intensity.

In some embodiments, depth mapping systems can use structured light techniques in which a known pattern of illumination is projected onto the scene. Depth can be calculated based on the deformation of the pattern in an image of the scene. In some embodiments, depth mapping systems use stereoscopic techniques, in which the parallax shift between two images captured at different locations is used to measure depth. In some embodiments, depth mapping systems can sense the times of flight of photons to and from points in the scene in order to measure the depth coordinates. In some embodiments, depth mapping systems control illumination and/or focus and can use various sorts of image processing techniques.

In the embodiment illustrated in FIG. 1, a user of the system 10, such as a healthcare professional 26 (for example, a surgeon performing the procedure), wears the head-mounted unit 28. In some embodiments, the head-mounted unit 28 includes one or more see-through displays 30, for example as described in the above-mentioned U.S. Pat. No. 9,928,629 or in the other patents and applications cited above.

In some embodiments, the one or more see-through displays 30 include an optical combiner. In some embodiments, the optical combiner is controlled by one or more processors 32. In some embodiments, the one or more processors 32 is disposed in a central processing system 50. In some embodiments, the one or more processors 32 is disposed in the head-mounted unit 28. In some embodiments, the one or more processors 32 are disposed in both the central processing system 50 and the head-mounted unit 28 and can share processing tasks and/or allocate processing tasks between the one or more processors 32.

In some embodiments, the one or more see-through displays 30 display an augmented-reality image to the healthcare professional 26. In some embodiments, the augmented reality image viewable through the one or more see-through displays 30 is a combination of objects visible in the real world with the computer-generated image. In some embodiments, each of the one or more see-through displays 30 comprises a first portion 33 and a second portion 35. In some embodiments, the one or more see-through displays 30 display the augmented-reality image such that the computer-generated image is projected onto the first portion 33 in alignment with the anatomy of the body of the patient 20 that is visible to the healthcare professional 26 through the second portion 35.

In some embodiments, the computer-generated image includes a virtual image of one or more tools 22. In some embodiments, the system 10 combines at least a portion of the virtual image of the one or more tools **22** into the computer-generated image. For example, some or all of the tool **22** may not be visible to the healthcare professional **26** because, for example, a portion of the tool **22** is hidden by the patient's anatomy (e.g., a distal end of the tool **22**). In some embodiments, the system **10** can display the virtual image of at least the hidden portion of the tool **22** as part of the computer-generated image displayed in the first portion **33**. In this way, the virtual image of the hidden portion of the tool **22** is displayed on the patient's anatomy. In some embodiments, the portion of the tool **22** hidden by the patient's anatomy increase and/or decreases over time or during the procedure. In some embodiments, the system **10** increase and/or decreases the portion of the tool **22** included in the computer-generated image based on the changes in the portion of the tool **22** hidden by the patient's anatomy over time.

Some embodiments of the system **10** comprise an anchoring device (e.g., bone marker **60**) for indicating the body of the patient **20**. For example, in image-guided surgery and other surgeries that utilize the system **10**, the bone marker **60** can be used as a fiducial marker. In some embodiments, the bone marker **60** can be coupled with the fiducial marker. In system **10**, for example, the anchoring device is configured as the bone marker **60** (e.g., anchoring device coupled with a marker that is used to register an ROI of the body of the patient **20**). In some embodiments, the anchoring device is coupled with a tracking system, with a preoperative or intraoperative CT scan of the ROI. During the procedure, in some embodiments, the tracking system (for example an IR tracking system) tracks the marker mounted on the anchoring device and the tool **22** mounted with a tool marker **40**. Following that, the display of the CT image data, including, for example, a model generated based on such data, on the near-eye display may be aligned with the surgeon's actual view of the ROI based on this registration. In addition, a virtual image of the tool **22** may be displayed on the CT model based on the tracking data and the registration. The user may then navigate the tool **22** based on the virtual display of the tool **22** with respect to the patient image data, and optionally, while it is aligned with the user view of the patient or ROI.

According to some aspects, the image presented on the one or more see-through displays **30** is aligned with the body of the patient **20**. According to some aspects, misalignment of the image presented on the one or more see-through displays **30** with the body of the patient **20** may be allowed. In some embodiments, the misalignment may be 0-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, 4-5 mm, 5-6 mm, and overlapping ranges therein. According to some aspects, the misalignment may typically not be more than about 5 mm. In order to account for such a limit on the misalignment of the patient's anatomy with the presented images, the position of the patient's body, or a portion thereof, with respect to the head-mounted unit **28** can be tracked. For example, in some embodiments, a patient marker **38** and/or the bone marker **60** attached to an anchoring implement or device such as a clamp **58** or pin, for example, may be used for this purpose, as described further hereinbelow.

When an image of the tool **22** is incorporated into the computer-generated image that is displayed on the head-mounted unit **28** or the HMD unit **70**, the position of the tool **22** with respect to the patient's anatomy should be accurately reflected. For this purpose, the position of the tool **22** or a portion thereof, such as the tool marker **40**, is tracked by the system **10**. In some embodiments, the system **10** determines the location of the tool **22** with respect to the patient's body such that errors in the determined location of the tool **22** with respect to the patient's body are reduced. For example, in certain embodiments, the errors may be 0-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, 4-5 mm, and overlapping ranges therein.

In some embodiments, the head-mounted unit **28** includes a tracking sensor **34** to facilitate determination of the location and orientation of the head-mounted unit **28** with respect to the patient's body and/or with respect to the tool **22**. In some embodiments, tracking sensor **34** can also be used in finding the position and orientation of the tool **22** and the clamp **58** with respect to the patient's body. In some embodiments, the tracking sensor **34** comprises an image-capturing device **36**, such as a camera, which captures images of the patient marker **38**, the bone marker **60**, and/or the tool marker **40**. For some applications, an inertial-measurement unit **44** is also disposed on the head-mounted unit **28** to sense movement of the user's head.

In some embodiments, the tracking sensor **34** includes a light source **42**. In some embodiments, the light source **42** is mounted on the head-mounted unit **28**. In some embodiments, the light source **42** irradiates the field of view of the image-capturing device **36** such that light reflects from the patient marker **38**, the bone marker **60**, and/or the tool marker **40** toward the image-capturing device **36**. In some embodiments, the image-capturing device **36** comprises a monochrome camera with a filter that passes only light in the wavelength band of light source **42**. For example, the light source **42** may be an infrared light source, and the camera may include a corresponding infrared filter. In some embodiments, the patient marker **38**, the bone marker **60**, and/or the tool marker **40** comprise patterns that enable a processor to compute their respective positions, i.e., their locations and their angular orientations, based on the appearance of the patterns in images captured by the image-capturing device **36**. Suitable designs of these markers and methods for computing their positions and orientations are described in the patents and patent applications incorporated herein and cited above.

In addition to or instead of the tracking sensor **34**, the head-mounted unit **28** can include a depth sensor **37**. In the embodiment shown in FIG. **2**A, the depth sensor **37** comprises a light source **46** and a camera **43**. In some embodiments, the light source **46** projects a pattern of structured light onto the region of interest (ROI) that is viewed through the one or more displays **30** by a user, such as professional **26**, who is wearing the head-mounted unit **28**. The camera **43** can capture an image of the pattern on the ROI and output the resulting depth data to the processor **32** and/or processor **45**. The depth data may comprise, for example, either raw image data or disparity values indicating the distortion of the pattern due to the varying depth of the ROI. In some embodiments, the processor **32** computes a depth map of the ROI based on the depth data generated by the camera **43**.

In some embodiments, the camera **43** also captures and outputs image data with respect to the markers in system **10**, such as patient marker **38**, bone marker **60**, and/or tool marker **40**. In this case, the camera **43** may also serve as a part of tracking sensor **34**, and a separate image-capturing device **36** may not be needed. For example, the processor **32** may identify patient marker **38**, bone marker **60**, and/or tool marker **40** in the images captured by camera **43**. The processor **32** may also find the 3D coordinates of the markers in the depth map of the ROI. Based on these 3D coordinates, the processor **32** is able to calculate the relative positions of the markers, for example in finding the position of the tool **22** relative to the body of the patient **20**, and can use this information in generating and updating the images presented on head-mounted unit **28**.

In some embodiments, the depth sensor **37** may apply other depth mapping technologies in generating the depth data. For example, the light source **46** may output pulsed or time-modulated light, and the camera **43** may be modified or replaced by a time-sensitive detector or detector array to measure the time of flight of the light to and from points in the ROI. As another option, the light source **46** may be replaced by another camera, and the processor **32** may compare the resulting images to those captured by the camera **43** in order to perform stereoscopic depth mapping. These and all other suitable alternative depth mapping technologies are considered to be within the scope of the present disclosure.

In the pictured embodiment, system **10** also includes a tomographic imaging device, such as an intraoperative computerized tomography (CT) scanner **41**. Alternatively or additionally, processing system **50** may access or otherwise receive tomographic data from other sources; and the CT scanner itself is not an essential part of the present system. In some embodiments, regardless of the source of the tomographic data, the processor **32** can computes a transformation over the ROI so as to register the tomographic images with the depth maps that it computes on the basis of the depth data provided by depth sensor **37**. The processor **32** can then apply this transformation in presenting a part of the tomographic image on the one or more displays **30** in registration with the ROI viewed through the one or more displays **30**. This functionality is described further hereinbelow with reference to FIG. **4**A.

In some embodiments, in order to generate and present an augmented reality image on the one or more displays **30**, the processor **32** computes the location and orientation of the head-mounted unit **28** with respect to a portion of the body of patient **20**, such as the patient's back. In some embodiments, the processor **32** also computes the location and orientation of the tool **22** with respect to the patient's body. In some embodiments, the processor **45**, which can be integrated within the head-mounted unit **28**, may perform these functions. Alternatively or additionally, the processor **32**, which is disposed externally to the head-mounted unit **28** and can be in wireless communication with the head-mounted unit **28**, may be used to perform these functions. The processor **32** can be part of the processing system **50**, which can include an output device **52**, for example a display, such as a monitor, for outputting information to an operator of the system, and/or an input device **54**, such as a pointing device, a keyboard, or a mouse, to allow the operator to input data into the system.

Alternatively or additionally, users of the system **10** may input instructions to the processing system **50** using a gesture-based interface. For this purpose, for example, the depth sensor **37** may sense movements of a hand **39** of the healthcare professional **26**. Different movements of the professional's hand and fingers may be used to invoke specific functions of the one or more displays **30** and of the system **10**.

In general, in the context of the present description, when a computer processor is described as performing certain steps, these steps may be performed by external computer processor **32** and/or computer processor **45** that is integrated within the head-mounted unit. The processor or processors carry out the described functionality under the control of suitable software, which may be downloaded to system **10** in electronic form, for example over a network, and/or stored on tangible, non-transitory computer-readable media, such as electronic, magnetic, or optical memory.

FIG. **2**B is a schematic pictorial illustration showing details of a head-mounted AR display (HMD) unit **70**, according to another embodiment of the disclosure. HMD unit **70** may be worn by the healthcare professional **26**, and may be used in place of the head-mounted unit **28** (FIG. **1**). HMD unit **70** comprises an optics housing **74** which incorporates a camera **78**, and in the specific embodiment shown, an infrared camera. In some embodiments, the housing **74** comprises an infrared-transparent window **75**, and within the housing, i.e., behind the window, are mounted one or more, for example two, infrared projectors **76**. One of the infrared projectors and the camera may be used, for example, in implementing a pattern-based depth sensor.

In some embodiments, mounted on housing **74** are a pair of augmented reality displays **72**, which allow professional **26** to view entities, such as part or all of patient **20**, through the displays, and which are also configured to present to surgeon images or any other information. In some embodiments, the displays **72** present planning and guidance information, as described above.

In some embodiments, the HMD unit **70** includes a processor **84**, mounted in a processor housing **86**, which operates elements of the HMD unit. In some embodiments, an antenna **88**, may be used for communication, for example with processor **32** (FIG. **1**).

In some embodiments, a flashlight **82** may be mounted on the front of HMD unit **70**. In some embodiments, the flashlight may project visible light onto objects so that professional is able to clearly see the objects through displays **72**. In some embodiments, elements of the HMD unit **70** are powered by a battery (not shown in the figure), which supplies power to the elements via a battery cable input **90**.

In some embodiments, the HMD unit **70** is held in place on the head of professional **26** by a head strap **80**, and the professional may adjust the head strap by an adjustment knob **92**.

In some embodiments, the HMD may comprise a visor, which includes an AR display positioned in front of each eye of the professional and controlled by the optical engine to project AR images into the pupil of the eye.

In some embodiments, the HMD may comprise a light source for tracking applications, comprising, for example, a pair of infrared (IR) LED projectors, configured to direct IR beams toward the body of patient **20**. In some embodiments, the light source may comprise any other suitable type of one or more light sources, configured to direct any suitable wavelength or band of wavelengths of light. The HMD may also comprise one or more cameras, for example, a red/green/blue (RGB) camera having an IR-pass filter, or a monochrome camera configured to operate in the IR wavelengths. The one or more camera s may be configured to capture images including the markers in system **10** (FIG. **1**).

In some embodiments, the HMD may also comprise one or more additional cameras, e.g., a pair of RGB cameras. In some embodiments, each RGB camera may be configured to produce high-resolution RGB (HR RGB) images of the patient's body, which can be presented on the AR displays. Because the RGB cameras are positioned at a known distance from one another, the processor can combine the images to produce a stereoscopic 3D image of the site being operated on.

In some embodiments, the HMD light source **46** (FIG. **2**A) may comprises a structured light projector (SLP) which projects a pattern onto an area of the body of patient **20** on which professional **26** is operating. In some embodiments, light source **46** comprises a laser dot pattern projector, which is configured to apply to the area structured light comprising a large number (typically between hundreds and hundreds of thousands) of dots arranged in a suitable pattern. This pattern serves as an artificial texture for identifying positions on large anatomical structures lacking fine details of their own, such as the skin and surfaces of the vertebrae. In some embodiments, one or more cameras **43** capture images of the pattern, and a processor, such as processor **32** (FIG. **1**), processes the images in order to produce a depth map of the area. In some embodiments, the depth map is calculated based on the local disparity of the images of the pattern relative to an undistorted reference pattern, together with the known offset between the light source and the camera.

In some embodiments, the projected pattern comprises a pseudorandom pattern of dots. In this case, clusters of dots can be uniquely identified and used for disparity measurements. In the present example, the disparity measurements may be used for calculating depth and for enhancing the precision of the 3D imaging of the area of the patient's body. In some embodiments, the wavelength of the pattern may be in the visible or the infrared range.

In some embodiments, the system **10** (FIG. **1**) may comprise a structured light projector (not shown) mounted on a wall or on an arm of the operating room. In such embodiments, a calibration process between the structured light projector and one or more cameras on the head-mounted unit or elsewhere in the operating room may be performed to obtain the 3D map.

FIG. **3**A is a schematic pictorial illustration showing details of the tool **22**, in accordance with an embodiment of the disclosure. In some embodiments, the tool **22** comprises a handle **162** and a shaft **164**. In some embodiments, the marker **40**, containing a predefined pattern, is disposed on the handle **162** in a fixed spatial relation to shaft **164**. Alternatively, marker **40** may protrude outward from tool **22** to ensure that it is visible to tracking sensor **34** and/or camera **43** (FIG. **2**A). Additionally or alternatively, the depth sensor **37** generates depth data with respect to tool **22**. Uses of the depth data in tracking and displaying images of tool **22** are described further hereinbelow with reference to FIG. **5**.

In some embodiments, based on the image information provided by tracking sensor **34** and/or camera **43**, processor **32** (or processor **45**) is able to compute the position (location and orientation) of marker **40** and to track changes in the position of the marker during the surgical procedure. In some embodiments, using the computed position and the known spatial relation of marker **40** to shaft **164**, processor **32** is thus able to find the orientation angle of a longitudinal axis **166** of shaft **164** and the location of a distal end **168** of the shaft, even when the shaft is inside the patient's body. On this basis, processor **32** generates an image of tool **22**, including shaft **164**, on displays **30** in registration with the ROI viewed through the displays.

In the example shown in FIG. **3**A, tool **22** is used in inserting a surgical implant **170** into the body of patient **20**. For the purpose of insertion, implant **170** is mounted on distal end **168** of shaft **164**. In one embodiment, the depth sensor **37** captures depth data with respect to both tool **22** and implant **170**. In some embodiments, the processor **32** uses the depth data in generating and displaying images of both the tool and the implant, as described further hereinbelow with reference to FIG. **6**.

FIG. **3**B is a schematic pictorial illustration showing details of tool **172**, in accordance with an alternative embodiment of the disclosure. Tool **172** is similar to tool **22** and includes similar components, which are labeled with the same indicator numbers. In contrast to tool **22**, however, tool **172** has a curved shaft **174**. The curvature of shaft **174** may be fixed, or it may be malleable or otherwise bendable, so that the shape of the shaft may vary from one operation to another or even during a single operation.

To identify the actual shape of shaft **174**, processor **32** receives and analyzes a depth map that includes tool **172**. The depth map may be captured using any suitable mapping and/or imaging technique, such as the various techniques described above. In some embodiments, the processor **32** segments the depth map to identify the 3D shape of shaft **174** and thus reconstructs a 3D model of tool **172**. Based on this 3D model, together with the location of tool marker **40**, in some embodiments, the processor **32** is able to compute the location of distal end **168** of shaft **174** even when the distal end is hidden from sight inside the patient's body. In some embodiments, the processor **32** can use the depth data in generating and displaying images of both tool **172** and implant **170**, which professional **26** inserts using tool **172**.

Methods for Image-Guided Surgery Using Depth Sensing

FIG. **4**A is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with an embodiment of the disclosure. This method, as well as the methods presented in the figures that follow, is described here with reference to the elements of system **10** (as shown in FIGS. **1**, **2**A and **3**A) for the sake of convenience and clarity. Alternatively, the principles of these methods may be applied, mutatis mutandis, in other sorts of augmented reality systems with suitable head-mounted units and depth sensing capabilities, including specifically the head-mounted units shown in FIG. **2**B, as well as in the system shown in FIG. **8**.

In some embodiments, the processing system **50** receives a 3D tomographic image, such as a CT image, of at least an ROI within the body, at an image input step **180**. This image may be based on a scan or scans performed before and/or in the course of the surgical procedure. In some embodiments during the procedure, depth sensor **37** generates depth data with respect to the ROI, and processor **32** applies the depth data in computing a depth map of the ROI, at a depth mapping step **182**. For example, the depth map may include the spine of patient **20**, which may be exposed in a surgical procedure.

In some embodiments, the processor **32** registers the 3D volume of the tomographic image from the CT scan with the depth map, at a registration step **184**. For this purpose, for example, processor **32** extracts 3D point clouds from both the tomographic image and the depth map and then computes a transformation to register one with the other. For instance, during surgery on the spine, the processor may compute the transformation by registering the spine in the depth map with the spine appearing in the tomographic image. Various methods for volume analysis may be used to perform the registration. In one embodiment, processor **32** computes triangular meshes based on the tomographic and depth data and then matches the features of the two meshes.

In some embodiments, the processor **32** applies the transformation computed at step **184** in presenting a part of the tomographic image on displays **30** of head-mounted unit **28** in registration with the ROI viewed through the displays, at a display step **186**. In accordance with several embodiments, to display a tracked tool together with the display of the ROI in step **186**, the depth sensor **37** (e.g., using cameras of the depth sensor **37**) must be calibrated with the tracking system (e.g., tracking sensor **34**). In some embodiments, the processor **32** may use the position of patient marker **38**, as detected by camera **43** and/or tracking sensor **34**, in improving the registration. For example, by identifying the position of patient marker **38** in the depth map, processor **32** may recognize where professional **26** is standing relative to the body of patient **20**. In some embodiments, the processor will then rotate the AR image that is presented on head-mounted unit **28** automatically according to the position of the head-mounted unit relative to the patient marker.

In some embodiments, the processor **32** continues to receive and process depth data output by depth sensor **37** during the surgical procedure. In some embodiments, the processor identifies changes in anatomical structures in the body of patient **20** during the procedure, based on the corresponding changes in the depth map. For example, processor **32** can detect that a part of a bone has been cut or that an implant has been inserted or removed. On this basis, processor **32** can modify the AR image presented on the display of head-mounted unit **28** to reflect the identified change.

FIG. **4**B is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with another embodiment of the disclosure. This embodiment builds on the method of FIG. **4**A and applies the method in detecting movement of the bone marker **60**, and/or movement of a bone anchoring implement or device such as the bone clamp **58** relative to the bone to which it is attached, such as the patient's spine. Identification and measurement of such movement is important since it can affect the overall accuracy of system **10**.

In some embodiments, the processor **32** identifies the location of bone marker **60** and/or the bone anchoring device in an initial depth map that is generated using depth sensor **37**, at a bone marker and/or bone anchoring device identification step **188**. In some embodiments, the depth map may be registered with the tomographic image of the patient's spine using the procedure of step **184** of FIG. **4**A, as described above. In some embodiments, the processor **32** can thus use the depth data in accurately calculating the position of bone marker **60** and/or bone clamp **58** relative the patient's spine.

Subsequently, in the course of the surgery in some embodiments, the depth sensor **37** continues to capture depth data, and processor periodically updates the depth map, at a depth update step **190**. In one embodiment, processor identifies bone marker **60** and/or bone clamp **58** in the updated depth map and checks whether the location of the clamp has shifted relative to the location of spine, as derived from the tomographic image, at a movement detection step **192**. When the location of the clamp in the updated depth map has changed, processor **32** takes a corrective action. For example, the processor **32** may modify the image presented on display **30** to reflect this change, at an image update step **194**. Additionally or alternatively, the processor **32** may issue a warning to the healthcare professional **26** that one of the markers has moved, such as the bone marker **60** or the patient marker **38**, so that the professional can take corrective action if needed.

FIG. **5** is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with yet another embodiment of the disclosure. In some embodiments, the method uses both the 3D shape of the tool **22**, which processor **32** derives from the depth data provided by depth sensor **37**, and the position of the tool marker **40**. The position of the tool marker **40** may be measured by tracking sensor **34**, or it may alternatively be extracted from images captured by camera **43** of the depth sensor **37**. In either case, in some embodiments, with all the elements used in finding the position of marker **40** and generating depth data with respect to tool **22** being fixed to the head-mounted unit **28** in a known spatial relation, the position of the marker will be consistently registered with the depth map.

In some embodiments, the method of FIG. **5** includes two calibration phases: The first phase includes performing initial calibration prior to the procedure. The second phase provides continuous calibration to update the initial calibration during the procedure. There may be two manners in which the initial calibration may be performed and then updated. In both manners, tool **22** and marker **40** (while the marker is mounted on the tool) are modeled by using depth sensing, for example to identify the trajectory and tip of the tool and the position of the marker with respect to the tool. In some embodiments, the processing system **50** then identifies the tool and marker and determines their relative locations. This function can be carried out using computer vision, for example using function of detection and segmentation, and optionally by applying machine learning techniques, including deep learning. Alternatively or additionally, tracking sensor **34** is used to identify the marker in the 3D model, in which case computer vision may be used only to identify the tool. In the continuous calibration update phase, either or both methods may be used, as well. Although the method shown in FIG. **5** and described below includes both calibration phases, aspects of the method may be applied only in the initial calibration without updating. Furthermore, although some steps in FIG. **5** and in the description below make use of tool marker **40**, the method may alternatively be carried out, mutatis mutandis, using only depth sensing without reliance on a tool marker.

In some embodiments, the processor **32** identifies tool marker **40** in an image of tool **22**, at a marker identification step **200**. As noted earlier, the image may be provided either by tracking sensor **34** or by camera **43**. In some embodiments, the processor **32** analyzes the image to derive the position of the tool marker.

In some embodiments, the processor **32** also processes depth data provided by depth sensor **37** in order to generate a depth map of tool **22**, possibly along with other elements in the ROI, at a depth mapping step **202**. In some embodiments, the processor **32** analyzes the shape of tool **22** in the depth map, for example using techniques of 3D image segmentation and/or machine learning that are known in the art. On this basis, for example, the processor **32** may identify shaft **164** (FIG. **3**A) and is thus able to find the orientation of axis **166**, corresponding to the "trajectory" of tool **22**, and the location of distal end or tip **168**. When the shaft of the tool is not straight, for example as in tool **172** (FIG. **3**B), processor **32** may use the information provided by the depth map in modeling the tool and finding the location and orientation of the distal tip.

Based on the marker position found at step **200** and the depth map analysis of step **202**, in some embodiments, the processor **32** computes a spatial transformation between the position of tool marker **40** and the location and orientation of shaft **164**, at a transformation step **204**. This transformation can be computed on the basis of the position of the marker and the depth data, without requiring prior knowledge of the shape of tool **22** or of the precise position of marker **40** relative to the other parts of the tool. Alternatively, a priori information regarding the position of the marker on the tool may be used to improve the accuracy of the transformation.

Using the tool marker **40** and/or updated depth data, the processor **32** tracks or may receive tracking information about the position of the tool **22** as professional **26** manipulates the tool with the shaft **164** inserted into the body (and thus not visible to depth sensor **37**), at a tracking step **206**. Using the tracked position and the spatial transformation computed at step **204**, processor **32** generates an image of tool **22** on display **30**, in registration with the ROI viewed through the display. The image specifically shows shaft **164**, and particularly distal end **168**, which when placed inside the patient's body may not be directly visible to professional **26**.

As noted earlier, although tool marker **40** is shown in FIG. **3** as a part of handle **162**, in other embodiments the tool marker may be fixed externally to the tool. In this case, the tool marker is liable to shift relative to the tool. Processor **32** analyzes the depth data provided by depth sensor **37** in order to calibrate the position of the tool marker relative to the tool. Thus, according to some embodiments, the processor may optionally update this analysis periodically at step **206** during the surgical procedure and checks the position to determine whether the tool marker has shifted relative to the tool, at a shift detection step **207**. If so, processor **32** recalibrates the tool position to compensate for the shift, and based on the new calibration, updates the transformation that was computed previously (at step **204**), at a transformation update step **208**. The image presented on display **30** is modified accordingly.

It may also occur that the shape of tool **22** or tool **172** changes in the course of the surgical procedure. For example, the bending angle of shaft **174** may change, or the shaft may even break. Processor **32** analyzes the depth data provided by depth sensor **37** in order to determine whether the shape of the tool has changed, at a shape change detection step **209**. If so, processor **32** recalibrates the tool shape to compensate for the change, and based on the new calibration, updates the transformation that was computed previously (at step **204**), at transformation update step **208**. The image of the tool on display **30** is updated, as well.

FIG. **6** is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with a further embodiment of the disclosure. This embodiment extends the method of FIG. **5** to deal with the shape and position of implant **170**, in addition to tool **22**. Although FIG. **3** shows implant **170** as a screw (for example a pedicle screw), the present method is equally applicable to implants of other types, such as stents, cages, and interbody devices. The present method is advantageous in that it enables processor **32** to extract the shapes of implants without requiring prior knowledge of the implant characteristics, and to use the extracted shapes in presenting virtual images of the implants on display **30**.

As an initial step, before inserting implant **170** into the body of patient **20**, processor **32** generates (or accesses if depth maps were previously generated) depth maps of both implant **170** and tool **22**, at a depth mapping step **210**. For this purpose, for example, a user of system **10** may be prompted to operate depth sensor **37** to capture depth data with respect to implant **170** by itself and also with respect to tool **22** with implant **170** mounted on distal end **168**. Processor **32** analyzes the depth data to identify the shape of implant **170**. Based on this shape and on the depth map of implant **170** mounted on tool **22**, processor **32** computes a spatial transformation between the position of the tip of tool **22** and the location and orientation of implant **170**, at a registration step **212**. A spatial transformation between the position of marker **40** and the location and orientation of implant **170** may be computed therefrom. Alternatively or additionally, a transformation between marker **40** and the tip of tool **22** may have been computed at an earlier phase, for example at a calibration phase as described above.

Following these steps, professional **26** uses tool **22** to insert implant **170** into the body of patient **20**. Processor **32** tracks the position of marker **40** as professional **26** manipulates shaft **164** of tool **22** within the body. According to some aspects, such tracking information is accessed by processor **32**. Using the tracked position, the spatial transformation computed at step **212**, and the shape of the implant that was identified at step **210**, processor **32** generates a virtual image on display **30** showing the implant within the body and/or its navigation, in registration with the ROI viewed through the display, at an image generation step **214**.

After implant **170** has been inserted in place, processor **32** may continuing processing the depth data provided by depth sensor **37** in order to identify and locate the implant, at an implant tracking step **216**. For example, processor **32** may identify the head of a screw in the depth map as a cue to the location and orientation of the screw. The display on head-mounted unit **28** can be modified to show the implant in the proper location. The information regarding the location and orientation of the implant can also be useful in planning subsequent stages of the procedure, such as insertion of rods between the screws.

FIG. **7** is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with an additional embodiment of the disclosure. This method is applicable particular in osteotomies, to assist in visualizing the portion of a bone that is removed in the procedure. In such a procedure, processing system **50** accesses a plan made by the surgeon to remove a certain volume of the bone in question, for example a part of one or more vertebrae, at a planning step **220**. The present method can assist the surgeon in comparing the volume of the bone that has been cut out to the planned volume.

For this purpose, prior to cutting of the bone, the processor **32** processes depth data generated by depth sensor **37** so as to identify the 3D shape of the bone, for example by generating a point cloud. The 3D cutting plan can be overlaid on the patient's actual anatomy (without displaying patient image data), optionally in a semi- or partially-transparent manner. For example, the top plane of the plan can be overlaid on the patient in alignment with and oriented according to the patient anatomy. The plan may be used as a guide for cutting.

As the surgeon cuts the bone, the outline or indication of a plane of the bone to be removed according to plan changes according to the cutting tool tip location, for example based on depth of penetration. The plan outline or plane can be displayed from a point of view defined by the tool orientation, as determined by tracking the tool tracking. This mode is especially compatible with a "Tip View" mode, in which the patient spine model displayed on the near-eye display, generated based on CT data, changes according to the tool tip location and such that the upper surface of the model is the upper plane defined by the tool orientation and tip location. For example, the model may be cut up to a plane orthogonal to the tool trajectory or longitudinal orientation such that only a portion of the model is displayed.

It is also possible to display or indicate what was done already or the portion of the procedure already performed (e.g., what portion of the bone was already cut and what portion of the bone is left to be cut). The already-cut portion may be indicated on the plan, for example in a different color, or augmented on the image or on the actual patient anatomy or the plan may be updated to show only the remaining bone, and may also show when a portion of the bone was not cut according to plan. Tracking of the cutting may be performed based on tool tip tracking or by depth sensing.

To track cutting using depth sensing, a first depth image of the bone is captured prior to cutting. During the cutting, additional depth images are captured. The capturing may be performed on user request or automatically, continuously or at predefined time interval. Each depth image is compared to the previous one, and processor **32** identifies whether a bone portion was removed or not, i.e., whether cutting was performed. When a difference in the bone volume is identified, the portion of bone that was removed may be indicated on or compared to the plan and displayed to the user. The depth sensor and the tracking system may be calibrated for this purpose. Alternatively, or additionally, the depth maps may be registered with the CT model, for example using feature matching. The calibration and registration process may allow comparison between the different depth maps, the CT model, and the plan.

After the bone has been cut, in some embodiments, the processor **32** accesses and processes new depth data in order to identify the modified 3D shape of the bone. Based on the difference between the 3D shapes, processor **32** identifies the portion of the bone that was removed, at an excision identification step **222**. In some embodiments, the processor **32** can then display an image showing the part of the bone that was removed in the surgical procedure, at a display step **224**. The surgeon can compare this image to the plan in order to verify that the osteotomy was completed according to plan. In some embodiments, the processor **32** can display both images, i.e., of the removed bone volume and of the planned volume, simultaneously to facilitate comparison between the two. The images may be displayed adjacent to or overlaid on one another.

Alternatively, processor **32** may display to the surgeon only the removed portion of the bone, without comparison to a plan. The processor may thus demonstrate the removed volume and assist in confirming the procedure or in deciding whether a correction or a further operation is required, for example. Additionally or alternatively, in cases in which an implant is to be placed in the body in place of the removed portion of the bone, the surgeon and/or processor **32** may use the model of the removed bone portion to select a suitable implant or to determine whether a particular implant is suitable. On this basis, a suitable implant may be selected from a database, for example. When comparing the removed bone volume to a specific implant, size data may be provided with respect to the implant, or it may be generated using the depth sensing techniques described above.

In some embodiments, the processor **32** generates a model of the removed portion of the bone, which can be used as an input for an implant printing device (such as a 3D printer). The implant may thus be designed based on the model of the removed portion of bone.

In one embodiment, the image and/or model of the ROI that is presented on the AR display, such as the patient spine model, dynamically changes according to the cutting performed and based on the tracking of cutting that is described above. Thus, if a drill is used, for example, then holes may be formed in the model correspondingly, based on tool tracking, for example. The model may be updated in this manner during the procedure and during cutting. At the end of the procedure, the user may be presented with a model showing the entire bone portion removed.

Alternative Embodiment

FIG. **8** is a schematic pictorial illustration of a system **240** for image-guided surgery, in accordance with an alternative embodiment of the invention. System **240** is similar in design and operation to system **10** (FIG. **1**), except that system **240** does not use the sorts of patient marker and clamp marker that were described above, with the accompanying tracking system. Rather, system **240** comprises an array of fiducial markers **242**, which are fixed to the back of patient **20**, for example using a suitable adhesive. Fiducial markers **242** comprises 3D elements, such as metal beads, which are both radiopaque and visible to cameras, such as camera **43** (FIG. **2**A).

Prior to the surgical procedure, markers **242** are fixed to the back of patient **20**, and a CT scan of the patient is performed, for example using CT scanner **41** (FIG. **1**). Markers **242** will appear in the resulting CT images along with the patient's skeleton. During the surgery, depth sensor **37** on head-mounted unit **28** captures depth maps of the patient's back, including markers **242**. Processor **32** analyzes the depth maps to find the 3D coordinates of markers **242**. The processor then matches the locations of markers **242** in the CT image data with the 3D coordinates of the markers in the depth map, and thus derives the appropriate transformation to register the CT image data with the actual patient anatomy viewed by professional **26**. Based on this registration, processor **32** presents parts of the CT images on displays **30**, overlaid on the patient anatomy.

The other features and applications of depth mapping in image-guided surgery that are described above may likewise be applied, mutatis mutandis, in system **240**.

Although the drawings and embodiments described above relate specifically to surgery on the spine, the principles of the present disclosure may similarly be applied in other sorts of surgical procedures, such as operations performed on the cranium and various joints, as well as dental surgery. It will thus be appreciated that the embodiments described above are cited by way of example, and that the present disclosure is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present disclosure includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.

Indeed, although the systems and processes have been disclosed in the context of certain embodiments and examples, it will be understood by those skilled in the art that the various embodiments of the systems and processes extend beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the systems and processes and obvious modifications and equivalents thereof. In addition, while several variations of the embodiments of the systems and processes have been shown and described in detail, other modifications, which are within the scope of this disclosure, will be readily apparent to those of skill in the art based upon this disclosure. It is also contemplated that various combinations or sub-combinations of the specific features and aspects of the embodiments may be made and still fall within the scope of the disclosure. It should be understood that various features and aspects of the disclosed embodiments can be combined with, or substituted for, one another in order to form varying modes of the embodiments of the disclosed systems and processes. Any methods disclosed herein need not be performed in the order recited. Thus, it is intended that the scope of the systems and processes herein disclosed should not be limited by the particular embodiments described above.

It will be appreciated that the systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible or required for the desirable attributes disclosed herein. The various features and processes described above may be used independently of one another or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of this disclosure.

Certain features that are described in this specification in the context of separate embodiments also may be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment also may be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination. No single feature or group of features is necessary or indispensable to each and every embodiment.

Each of the processes, methods, and algorithms described in the preceding sections may be embodied in, and fully or partially automated by, code modules executed by one or more computer systems or computer processors comprising computer hardware. The code modules may be stored on any type of non-transitory computer-readable medium or computer storage device, such as hard drives, solid state memory, optical disc, and/or the like. The systems and modules may also be transmitted as generated data signals (for example, as part of a carrier wave or other analog or digital propagated signal) on a variety of computer-readable transmission mediums, including wireless-based and wired/cable-based mediums, and may take a variety of forms (for example, as part of a single or multiplexed analog signal, or as multiple discrete digital packets or frames). The processes and algorithms may be implemented partially or wholly in application-specific circuitry. The results of the disclosed processes and process steps may be stored, persistently or otherwise, in any type of non-transitory computer storage such as, for example, volatile or non-volatile storage.

The various features and processes described above may be used independently of one another or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of this disclosure. In addition, certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be performed in other sequences that are appropriate. For example, described blocks or states may be performed in an order other than that specifically disclosed, or multiple blocks or states may be combined in a single block or state. The example blocks or states may be performed in serial, in parallel, or in some other manner. Blocks or states may be added to or removed from the disclosed example embodiments. The example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed example embodiments.

Conditional language, such as, among others, "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

The terms "comprising," "including," "having," and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. In addition, the term "or" is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term "or" means one, some, or all of the elements in the list. In addition, the articles "a," "an," and "the" as used in this application and the appended claims are to be construed to mean "one or more" or "at least one" unless specified otherwise. Similarly, while operations may be depicted in the drawings in a particular order, it is to be recognized that such operations need not be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Further, the drawings may schematically depict one or more example processes in the form of a flowchart. However, other operations that are not depicted may be incorporated in the example methods and processes that are schematically illustrated. For example, one or more additional operations may be performed before, after, simultaneously, or between any of the illustrated operations. Additionally, the operations may be rearranged or reordered in other embodiments. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems may generally be integrated together in a single software product or packaged into multiple software products. Additionally, other embodiments are within the scope of the following claims. In some cases, the actions recited in the claims may be performed in a different order and still achieve desirable results.

As used herein "generate" or "generating" may include specific algorithms for creating information based on or using other input information. Generating may include retrieving the input information such as from memory or as provided input parameters to the hardware performing the generating. Once obtained, the generating may include combining the input information. The combination may be performed through specific circuitry configured to provide an output indicating the result of the generating. The combination may be dynamically performed such as through dynamic selection of execution paths based on, for example, the input information, device operational characteristics (for example, hardware resources available, power level, power source, memory levels, network connectivity, bandwidth, and the like). Generating may also include storing the generated information in a memory location. The memory location may be identified as part of the request message that initiates the generating. In some implementations, the generating may return location information identifying where the generated information can be accessed. The location information may include a memory location, network locate, file system location, or the like.

Any process descriptions, elements, or blocks in the flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process. Alternate implementations are included within the scope of the embodiments described herein in which elements or functions may be deleted, executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those skilled in the art.

All of the methods and processes described above may be embodied in, and partially or fully automated via, software code modules executed by one or more general purpose computers. For example, the methods described herein may be performed by the processors **32**, **45** described herein and/or any other suitable computing device. The methods may be executed on the computing devices in response to execution of software instructions or other executable code read from a tangible computer readable medium. A tangible computer readable medium is a data storage device that can store data that is readable by a computer system. Examples of computer readable mediums include read-only memory, random-access memory, other volatile or non-volatile memory devices, CD-ROMs, magnetic tape, flash drives, and optical data storage devices.

Documents incorporated by reference in the present patent application are to be considered an integral part of the application except that, to the extent that any terms are defined in these incorporated documents in a manner that conflicts with definitions made explicitly or implicitly in the present specification, only the definitions in the present specification should be considered.

It should be emphasized that many variations and modifications may be made to the above-described embodiments, the elements of which are to be understood as being among other acceptable examples. All such modifications and variations are intended to be included herein within the scope of this disclosure. The foregoing description details certain embodiments. It will be appreciated, however, that no matter how detailed the foregoing appears in text, the systems and methods can be practiced in many ways. As it is also stated above, it should be noted that the use of particular terminology when describing certain features or aspects of the systems and methods should not be taken to imply that the terminology is being re-defined herein to be restricted to including any specific characteristics of the features or aspects of the systems and methods with which that terminology is associated. While the embodiments provide various features, examples, screen displays, user interface features, and analyses, it is recognized that other embodiments may be used.

What is claimed is:

1. A head-mounted system for image-guided surgery, comprising:

a head-mounted unit, comprising a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a body of a patient that is viewed through the display by a user wearing the head-mounted unit, the depth data including data for both a region of interest (ROI) within the body of the patient and an array of radiopaque fiducial markers fixed externally to the body of the patient; and

a processor, which is configured to:

receive a three-dimensional (3D) tomographic image of the body of the patient;

compute a depth map based on the depth data generated by the depth sensor, the depth map including both the ROI within the body of the patient and the array of radiopaque fiducial markers fixed externally to the body of the patient;

compute a transformation over the ROI so as to register the tomographic image with the depth map; and

apply the transformation in presenting a part of the tomographic image on the display in registration with the ROI viewed through the display,

wherein the ROI within the body of the patient that is included in the depth map includes a spine of the patient, which is exposed in a surgical procedure,

wherein the processor is configured to compute the transformation by registering the spine in the depth map with the spine appearing in the tomographic image,

wherein the tomographic image comprises a CT scan of the patient which was performed with the array of radiopaque fiducial markers fixed externally to the body of the patient,

wherein the processor is configured to identify respective 3D coordinates of the fiducial markers in the depth map and to register the CT scan with the ROI viewed through the display by matching the fiducial markers in the CT to the respective 3D coordinates, and

wherein the processor is configured to compute an updated depth map in a course of a surgery to enable detection of a change in the ROI within the body of the patient based on the updated depth map.

2. The system of claim **1**, wherein the ROI comprises a bone of the body to which an anchoring device is coupled, and wherein the processor is further configured to:

identify a location of the anchoring device in the depth map;

detect a change in the location of the anchoring device in the updated depth map; and

take a corrective action responsively to the change.

3. The system of claim **2**, wherein the corrective action comprises modifying a presentation on the display responsively to the change in the location of the anchoring device.

4. The system of claim **1**, wherein the processor is configured to:

process the updated depth map so as to identify a change in an anatomical structure in the body of the patient during the surgery; and

modify the image presented on the display responsively to the identified change.

5. The system of claim **1**, wherein the processor is configured to:

process the updated depth map so as to identify an implant inserted into the body of the patient during the surgery; and

modify the image presented on the display responsively to the identified implant.

6. The system of claim **1**, wherein the 3D tomographic image is a computed tomographic image.

7. The system of claim **1**, wherein the depth sensor comprises a pattern projector and a camera configured for structured-light depth mapping.

8. The system of claim **7**, wherein the pattern projector comprises a laser dot pattern projector configured to apply structured light comprising dots arranged in a pattern.

9. The system of claim **8**, wherein the pattern is a pseudorandom pattern of dots.

10. The system of claim **9**, wherein the depth map is calculated based on a local disparity of images of the pattern captured by the camera relative to an undistorted reference pattern, together with a known offset between the pattern projector and the camera.

11. The system of claim **1**, wherein the depth sensor comprises a pair of cameras configured for stereoscopic depth mapping.

12. The system of claim **1**, wherein the depth sensor comprises a beam projector and one or more detectors configured for time-of-flight measurement.

13. The system of claim **1**, wherein the head-mounted unit comprises a head strap.

14. The system of claim **1**, wherein the head-mounted unit comprises a visor.

15. The system of claim **1**, wherein the processor is further configured to:

process the depth map to identify a 3D shape of a vertebrae bone prior to a cutting of the vertebrae bone;

process the updated depth map to identify a modified 3D shape of the vertebrae bone after the cutting of the vertebrae bone;

based on a difference between the 3D shape and the modified 3D shape, identify a portion of the vertebrae bone that was removed; and

present an image on the display showing the identified portion of the vertebrae bone that was removed.

16. A head-mounted system for image-guided surgery comprising:

a see-through augmented-reality display disposed so as to be viewable by a user over a body of a patient;

a depth sensor configured to generate depth data with respect to the body of the patient, the depth data including data for both a region of interest (ROI) within the body of the patient and one or more markers positioned external to the body of the patient; and

a processor and a memory for storing instructions that, when executed by the processor cause the system to:

receive a three-dimensional (3D) tomographic image of the body of the patient;

determine a depth map based at least in part on the depth data, the depth map including both the ROI within the body of the patient and the one or more markers positioned external to the body of the patient;

determine a transformation over the ROI so as to register the 3D tomographic image with the depth map;

display at least a part of the 3D tomographic image on the see-through augmented-reality display in registration with the ROI based at least in part on the transformation; and

determine an updated depth map in a course of a surgery to enable detection of a change in the ROI within the body of the patient based on the updated depth map.

17. The system of claim **16**, wherein:

the 3D tomographic image is a computed tomographic image;

the depth sensor comprises a pattern projector and a camera configured for structured-light depth mapping; and

the pattern projector comprises a laser dot pattern projector configured to apply structured light comprising dots arranged in a pattern.

18. The system of claim **17**, wherein the pattern is a pseudorandom pattern of dots.

19. The system of claim **18**, wherein the depth map is calculated based on a local disparity of images of the pattern captured by the camera relative to an undistorted reference pattern, together with a known offset between the pattern projector and the camera.

20. The system of claim **16**, further comprising a head-mounted unit comprising the see-through augmented-reality display.

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